Aaron Espinoza

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EXPERIENCE

Open Source Contributor **Sept. 2025 – Present**
The FreeBSD Project *Remote*

- Fixed a kernel panic in `if_bridge(4)` caused by a NULL pointer dereference on an error path; found while tracing the driver during `veb(4)` work and landed through the project's code review process. Commit history
- Building `veb(4)`, a virtual Ethernet bridge pseudo-device inspired by OpenBSD's veb(4). It is a stripped-down L2 bridging driver aimed at virtual topologies that stays hidden from the host network stack, cutting latency for jail and bhyve networking. Out-of-tree module build with CI running load and jail networking tests.
- Maintain RustKLD, the Rust kernel driver framework delivered during GSoC, including CI and contributor docs.

Google Summer of Code Contributor **May 2025 – Sept. 2025**
The FreeBSD Project *Remote*

- Delivered a modular Rust framework for FreeBSD kernel device drivers with safe wrappers over C kernel objects, removing the need for contributors to hand-roll FFI for each new driver.
- Wrote 11 automated ATF/Kyua tests covering stress, concurrency, error handling, and memory leaks for existing drivers.
- Fixed devfs registration, quiesce handling, and LLVM 19 compatibility in the existing Rust driver; shipped with reference drivers and documentation to assist new contributors.

Undergraduate Deep Learning Researcher **Aug. 2024 – Present**
Texas State University *San Marcos, TX*

- Research GAN-based adversarial perturbation generation against facial recognition systems using hybrid U-Net generators.
- Achieved 98.7% average impersonation success on LFW and CASIA-WebFace while keeping perturbations visually imperceptible; designed both targeted and untargeted attacks to measure model resilience.

AI Research Intern **May 2024 – Aug. 2024**
Guangdong Provincial People's Hospital *Guangzhou, China*

- Placed 1st globally in the MICCAI 2024 WHS++ whole heart segmentation challenge.
- Built ensembled PyTorch CNN segmentation models for cardiac CT and MRI, reaching a 94% Dice score.
- Applied data augmentation and model calibration to hold accuracy across imaging modalities and hospital sites.

Maintenance Technician **Aug. 2022 – May 2024**
tekRESCUE *San Marcos, TX*

- Ran weekly maintenance across client server fleets; built custom servers and handled on-site troubleshooting.

PUBLICATIONS

Enhance Multi-modal and Multi-center Whole Heart Segmentation Using Data Augmentation and Model Calibration. *CARE Workshop, MICCAI 2024.* doi.org/10.1007/978-3-031-87009-5_13

PROJECTS

RustKLD github.com/Acesp25/rustkld
Framework and CI harness for writing Rust FreeBSD kernel device drivers.
Rust, C, FreeBSD, ATF/Kyua.

Portfolio Website acespinoza.net
Personal site written entirely in Rust. Templates are typed at compile time and CSS is minified into a single static binary.
Rust, Rocket, Maud.

EDUCATION

Texas State University **Expected Dec. 2026**
B.S. Honors Computer Science; Minors in Applied Mathematics and Chinese *San Marcos, TX*

- GPA 3.98/4.00 overall, 4.00/4.00 College of Science & Engineering GPA. Dean's List/President's List, all semesters.
- Extracurriculars: Vice President, TXST Fencing Club (40+ members), TXST Esports Club CS2 Captain.

SKILLS

Languages: C, Rust, Python, Java, C++, Bash, Assembly (MIPS, x86-64); Chinese (minor, conversational)
Systems & Networking: FreeBSD kernel and device drivers, POSIX (pthreads, sockets, IPC), pf, WireGuard, NFS, bhyve, QEMU, jails
Machine Learning: PyTorch, TensorFlow, CNNs, U-Nets, GANs
Tools: Git, ATF/Kyua, DTrace, GDB, Wireshark, Vim, VS Code, Jira